

Spectrum Sensing via Neural Operators

Red triad: "Problem $\rightarrow$ Solution $\rightarrow$ Optimization"

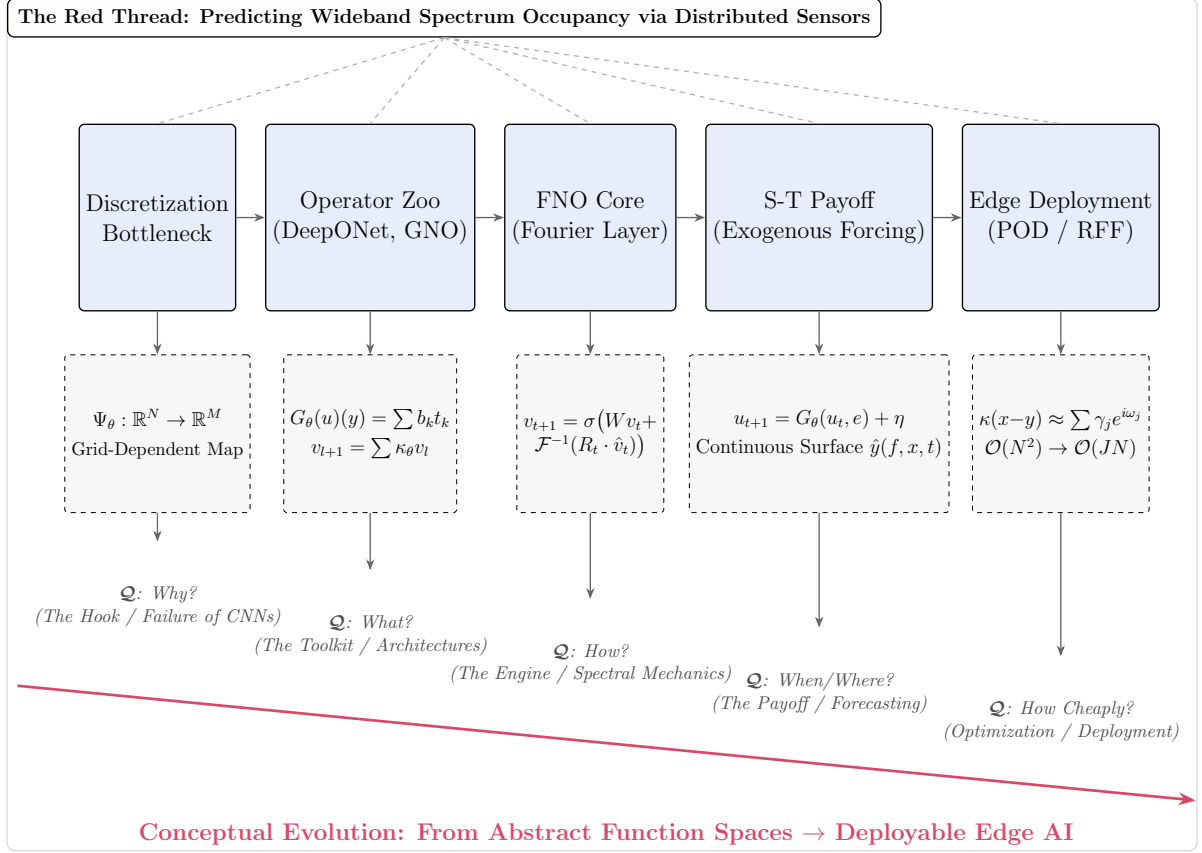


Figure 1: Conceptual evolution of the Neural Operator curriculum.

1 Discretization-Dependent Paradigms and Neural Operators

The Continuous Nature of Spectrum Data

Let $\Omega \subset \mathbb{R}^d$ be a compact spatial-frequency domain (e.g., time–frequency space). We define the input function space

$$\mathcal{U} = C(\Omega; \mathbb{R}^{d_u})$$

and the output function space

$$\mathcal{V} = C(\Omega'; \mathbb{R}^{d_v}),$$

both equipped with the supremum norm $\|\cdot\|_\infty$, rendering them Banach spaces. The physical quantity of interest, such as the received signal power or spectral occupancy, is a continuous function $u \in \mathcal{U}$.

Formal Definition of Grid-Dependence

Classical deep learning architectures, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), define parametric maps

$$\Psi_\theta : \mathbb{R}^{N \times d_u} \rightarrow \mathbb{R}^{M \times d_v},$$

where N and M denote the cardinalities of the discretization grids $\mathcal{X}_N \subset \Omega$ and $\mathcal{Y}_M \subset \Omega'$, respectively.

Definition 1 (Discretization-Dependence). A parametric map Ψ_θ is *grid-dependent* if varying the sampling resolution (i.e., $N \rightarrow N'$ or $M \rightarrow M'$) alters the input or output tensor dimensionality, rendering the architecture invalid on the new grid without complete architectural reconfiguration and retraining.

In the context of spectrum sensing, a CNN trained on data sampled at 100 MHz is inoperable when the sampling rate changes to 200 MHz, despite the underlying physical propagation laws remaining identical.

The Neural Operator as a Discretization-Invariant Alternative

Definition 2 (Neural Operator). A Neural Operator is a parametric map

$$G_\theta : \mathcal{U} \rightarrow \mathcal{V}$$

that approximates an underlying continuous operator G . The map G_θ is *discretization-invariant*: for any two grids $\mathcal{X}_N$ and $\mathcal{X}_{N'}$, the evaluations $G_\theta(u)|_{\mathcal{X}_N}$ and $G_\theta(u)|_{\mathcal{X}_{N'}}$ converge uniformly to the same continuous function $G(u)$ as $N, N' \rightarrow \infty$.

Remark 1. This discretization-invariance is the foundational prerequisite for learning the physical laws of electromagnetic propagation and fading, which exist independently of the sensor’s sampling frequency.

2 Foundational Neural Operator Architectures

This module presents three distinct architectural paradigms for implementing the Neural Operator defined in Module I, positioning the Fourier Neural Operator (FNO) as a specialized case.

Architecture II.A: DeepONet (Branch-Trunk Decomposition)

The DeepONet approximates the operator via a finite-sum factorization.

Definition 3 (DeepONet). Let $\mathcal{P} : \mathcal{U} \rightarrow \mathbb{R}^m$ be a finite-dimensional projection (e.g., point-wise evaluations at fixed sensor locations). The DeepONet is defined as

$$G_\theta(u)(y) = \sum_{k=1}^p b_k(\mathcal{P}(u)) \cdot t_k(y),$$

where $b_k : \mathbb{R}^m \rightarrow \mathbb{R}$ constitutes the *branch net* encoding the input function, and $t_k : \Omega' \rightarrow \mathbb{R}$ constitutes the *trunk net* encoding the query coordinates $y \in \Omega'$.

Example 1 (Spectrum Sensing Context). The DeepONet predicts the Received Signal Strength Indicator (RSSI) at a new GPS coordinate y not observed during training, by querying the trunk net at that coordinate.

Architecture II.B: Graph Neural Operator (GNO)

Let $\{x_i\}_{i=1}^N \subset \Omega$ denote irregularly distributed sensor locations. Define a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertices $\mathcal{V} = \{x_i\}$ and edges determined by a connectivity radius $r > 0$.

Definition 4 (Graph Neural Operator). The GNO iteratively updates vertex features via the integral-type aggregation:

$$v_{l+1}(x_i) = \sigma \left(\sum_{x_j \in \mathcal{N}(x_i)} \kappa_\theta(x_i, x_j) v_l(x_j) \right),$$

where $\mathcal{N}(x_i)$ is the spatial neighborhood, $\kappa_\theta : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^{d_v \times d_v}$ is a learnable kernel parametrized by a neural network, and σ is a point-wise non-linear activation.

Example 2 (Spectrum Sensing Context). GNO naturally accommodates the non-Euclidean, irregular placement of spectrum sensors across a metropolitan area.

Architecture II.C: Fourier Neural Operator (Conceptual Prelude)

In contrast to the local aggregation of GNO, the FNO exploits translational symmetry. The integral kernel is constrained to satisfy

$$\kappa(x, y) = \kappa(x - y).$$

This structural constraint permits the use of global convolutions and the Fourier transform, enabling a global receptive field in a single computational step. The rigorous derivation is deferred to Module III.

3 Fourier Neural Operators: Spectral Kernel Mechanics and Layer-wise Formulation

This module provides the rigorous mathematical derivation of the Fourier Layer.

The Spectral Kernel Operator

Consider the general integral operator acting on a function $v \in \mathcal{U}$:

$$(K(v))(x) = \int_{\Omega} \kappa(x, y) v(y) dy, \quad \forall x \in \Omega.$$

Theorem 1 (Convolution Theorem for Translation-Invariant Kernels). *If $\Omega = \mathbb{T}^d$ (the d -dimensional torus) and the kernel is translation-invariant, i.e., $\kappa(x, y) = \kappa(x - y)$, then the integral operator is a convolution. Applying the Convolution Theorem yields:*

$$(K(v))(x) = \mathcal{F}^{-1} (\mathcal{F}(\kappa) \cdot \mathcal{F}(v)) (x) = \mathcal{F}^{-1} (R(\xi) \cdot \hat{v}(\xi)) (x),$$

where $\mathcal{F}$ denotes the Fourier transform, $\xi \in \mathbb{Z}^d$ is the frequency/wavenumber vector, $\hat{v} = \mathcal{F}(v)$, and

$$R(\xi) = \mathcal{F}(\kappa)(\xi) \in \mathbb{C}^{d_v \times d_v}$$

is a learnable complex-valued weight tensor.

Frequency Decoupling and Truncation

To control computational complexity, we restrict the spectral representation to the lowest $k_{\max}$ Fourier modes, where $k_{\max} \ll N$. This truncation acts as a low-pass filter on the feature channels.

Remark 2. In spectrum sensing, low-frequency components correspond to global, slowly varying interference patterns (e.g., diurnal usage cycles). High-frequency components correspond to localized, rapid Rayleigh fading or burst noise. The FNO learns to weight these components independently.

The Complete Fourier Layer Formulation

Let $v_t \in \mathbb{R}^{N \times d_v}$ be the feature representation at layer t .

Definition 5 (Fourier Layer). The complete Fourier Layer mapping $v_t \mapsto v_{t+1}$ is formally defined as:

$$v_{t+1}(x) = \sigma \left(W v_t(x) + \mathcal{F}^{-1} (R_t(\xi) \cdot \mathcal{F}(v_t)(\xi)) \right),$$

where:

- $W \in \mathbb{R}^{d_v \times d_v}$ is a local linear transformation acting as a residual/skip connection;
- $R_t(\xi) \in \mathbb{C}^{d_v \times d_v}$ is the learnable spectral filter specific to layer t ;
- σ is a point-wise non-linear activation function (e.g., GELU).

Proposition 1 (Global Receptive Field). *The receptive field of the Fourier Layer is globally infinite. The layer index t corresponds exclusively to the successive nonlinear refinement of the feature manifold, not to an expansion of the spatial receptive field. This fundamentally differentiates FNOs from standard CNNs.*

4 Autoregressive Spatio-Temporal Forecasting with Exogenous Forcing

This module extends the deterministic FNO engine to address real-world stochastic dynamical systems.

Autoregressive Formulation of Stochastic Systems

Let $u(t, x) \in \mathbb{R}^{d_v}$ denote the state of the spectrum at time $t \in \mathbb{R}_{\geq 0}$ and spatial coordinate $x \in \Omega$.

Definition 6 (Autoregressive FNO). For a history length T , the state evolution is modeled as:

$$u(t + \Delta t) = G_\theta(u(t), u(t - \Delta t), \dots, u(t - (T - 1)\Delta t)) + \eta(t),$$

where $\eta(t) \sim \mathcal{N}(0, \Sigma)$ is a stochastic forcing term representing environmental noise, and G_θ is the FNO. The input tensor is structured as

$$\mathcal{X} \in \mathbb{R}^{B \times T \times N \times d_v},$$

with B denoting the batch size and N the spatial/frequency resolution.

Exogenous Forcing via Augmented Input Spaces

Define an exogenous variable $e(x, t) \in \mathbb{R}^{d_e}$ representing non-evolutionary external context (e.g., diurnal temporal encoding, weather conditions, public holiday indicators).

Definition 7 (Augmented State). The augmented input function is constructed via concatenation:

$$u_{\text{aug}}(x, t) = [u(x, t), e(x, t)] \in \mathbb{R}^{d_v+d_e}.$$

This augmented function is passed through the lifting layer $P : \mathbb{R}^{d_v+d_e} \rightarrow \mathbb{R}^{d_v}$ (as introduced in the Fourier Layer formulation of Module III).

Example 3 (Spectrum Sensing Context). By concatenating the hour-of-the-day as an exogenous channel, the FNO learns distinct regression surfaces for weekday afternoon traffic (high WiFi usage) versus late-night idle periods. This conditioning is formalized as learning a family of operators $G_\theta(\cdot|e)$.

Multidimensional Continuous Regression Surfaces

The final output is a continuous scalar field $\hat{y} \in \mathcal{V}$, representing the predicted SNR or occupancy probability.

$$\hat{y}(f, x_{\text{sensor}}, t) = \text{Proj}(G_\theta(u_{\text{aug}})),$$

where $\text{Proj} : \mathbb{R}^{d_v} \rightarrow \mathbb{R}$ projects the hidden feature dimension to a scalar value, and f denotes the frequency dimension.

Remark 3. This formulation yields smooth interpolation between discrete frequency bins and sensor locations, a property rigorously unattainable by conventional grid-based architectures due to their finite-dimensional parameterization.

5 Computational Burden Reduction: Dimensionality Reduction and Spectral Sampling

This module formalizes strategies to deploy the FNO on battery-powered, resource-constrained edge sensors (USRPs/SDRs). These are strictly *computational optimization hacks* applied to the core FNO, not architectural generalizations.

Proper Orthogonal Decomposition (POD) / Principal Component Analysis

Let $\{u^{(i)}\}_{i=1}^M \subset \mathcal{U}$ be a set of spectrum snapshots.

Definition 8 (POD-NO). Define the correlation matrix $C \in \mathbb{R}^{N \times N}$ with entries:

$$C_{ij} = \frac{1}{M} \sum_{m=1}^M u^{(m)}(x_i) u^{(m)}(x_j).$$

Solving the eigenvalue problem $C\phi_i = \lambda_i\phi_i$ yields an orthogonal basis $\{\phi_i\}_{i=1}^N$ sorted by descending eigenvalues λ_i . The POD-NO approximation of rank $r \ll N$ is:

$$u(x) \approx \sum_{i=1}^r a_i \phi_i(x), \quad a_i = \langle u, \phi_i \rangle_{\mathcal{U}}.$$

The FNO is applied to the reduced coefficient vector $a \in \mathbb{R}^r$ rather than the raw N -dimensional grid, reducing the FFT complexity from $\mathcal{O}(N \log N)$ to $\mathcal{O}(r \log r)$.

Bochner Sampling / Random Fourier Features (RFF)

Leverage Bochner’s theorem, which provides a spectral representation for translation-invariant kernels.

Theorem 2 (Bochner’s Theorem). *A continuous, positive-definite, translation-invariant kernel $k(x - y)$ on $\mathbb{R}^d$ is the Fourier transform of a positive finite measure $p(\xi)$:*

$$k(x - y) = \int_{\mathbb{R}^d} e^{i\xi \cdot (x-y)} dp(\xi).$$

Definition 9 (Random Fourier Features Approximation). Approximating the spectral integral via Monte Carlo sampling with J features yields:

$$k(x - y) \approx \sum_{j=1}^J \gamma_j e^{i\omega_j \cdot (x-y)}, \quad \omega_j \sim p(\xi),$$

where $\gamma_j \in \mathbb{C}$ are learnable coefficients.

Proposition 2 (Formal Complexity Reduction). *Replacing the full spectral matrix $R \in \mathbb{C}^{N \times N}$ (from Module III) with the RFF approximation reduces the memory complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(JN)$, where $J \ll N$.*

Remark 4. This approximation is strictly a *linear algebra optimization*. It preserves the global receptive field of the FNO while drastically reducing the multiply-accumulate (MAC) operations required for inference, enabling real-time deployment on edge spectrum sensors.